

Practice Problems

1. Construct an injective homomorphism from S_n inside $GL_n(\mathbb{R})$.
2. Show that if G is a group, then the map $g \mapsto g^{-1}$ is a homomorphism iff G is abelian. Similarly $g \mapsto g^2$ is a homomorphism iff G is abelian.
3. Show that $\mathbb{Z}$ is not isomorphic to $\mathbb{Q}$.
4. Show that $(\mathbb{Z}_{16})^*$ is not cyclic.
5. If $f : G \rightarrow G'$ is a group homomorphism and $N' \subset G'$ is a normal subgroup, show that $N = f^{-1}(N')$ is a normal subgroup of G .
6. Prove that intersection of two normal subgroups is again normal.
7. Consider the 3-cycle (123) in S_5 . Let H be the cyclic subgroup of (order 3) generated by it. Show that $H \subsetneq N_G(H)$.
8. Prove that the diagonal subgroup $D \subset S_3 \times S_3$ is not a normal subgroup.
9. Check rigorously that conjugation is a group action on a group.
10. Consider the action of $\mathbb{R}^*$ on $\mathbb{R}^2 - \{0\}$ by $\lambda(v, w) = (\lambda v, \lambda^{-1}w)$. Is this action free ? faithful ? transitive ?
11. Classify all abelian groups of order 27 up to isomorphism.